Wind energy
and wind turbines

Rafael Nogueira Nakashima

Department of Energy Conversion and Storage
Technical University of Denmark

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Outline

The wind resource
- Wind geography
- Temporal variability

Break

Wind turbine technology
- Technology fundamentals
- Wind turbine parts

Feedback

Activities

Learning objectives

Wind energy

By the end of this lecture, you should be able to

- Identify true or false statements related to wind energy resource, wind energy conversion, and wind turbine parts.
- Calculate technical parameters related to wind energy potential, wind energy conversion, and wind turbines.
- Calculate cross-disciplinary parameters related to capacity factor of wind energy.

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Origin of wind

pressure differences from temperature gradients

At large scales, wind is caused by uneven solar irradiance on Earth.

- Higher irradiation raises air temperature and creates low-pressure regions

$$\uparrow \bar{I} \implies \underbrace{\uparrow T \implies \downarrow \rho}_{\text{Air lifted}} \implies \underbrace{\downarrow P}_{\text{Air left}}$$

- Air moves to the low-pressure regions, forming winds

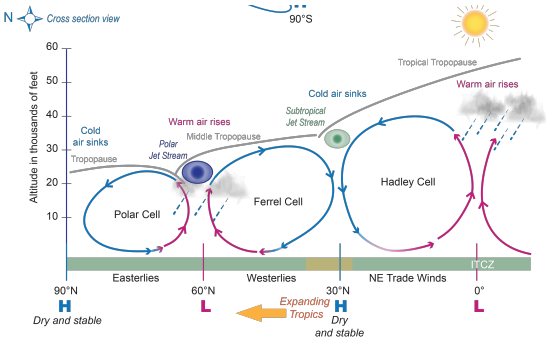


Figure: Cross-section view of the atmosphere circulation model [1]

Global scale wind

atmosphere circulation cells + Earth's rotation

- Air convection creates atmospheric air circulation cells (i.e., Hadley, Ferrel, and Polar)
- The Earth's rotation creates a drag and torque effect on atmospheric circulation.
- The combined effect of atmosphere circulation and Earth's rotation leads to prevailing wind directions.

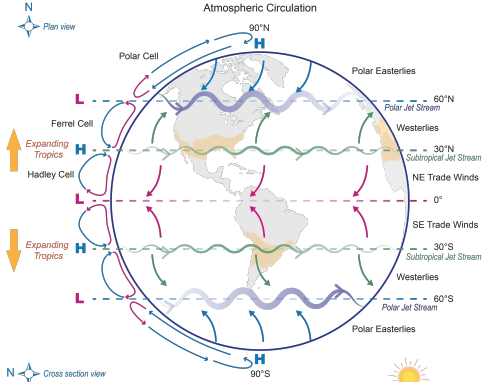


Figure: Coriolis effect on wind [1]

Local scale winds

sea/land and mountain breeze

Similarly, the winds can also be produced due to local differences in surface temperature

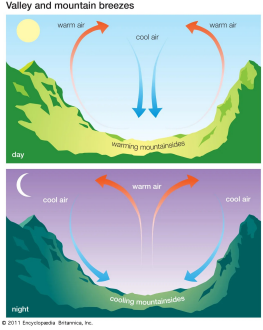


Figure: Mountain breeze [2]

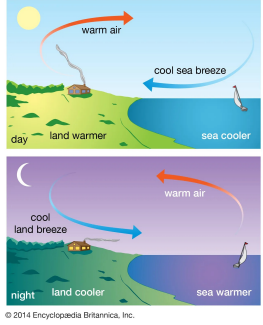


Figure: Sea/Land breezes cycle [2]

Geographical distribution

patterns in global wind atlas

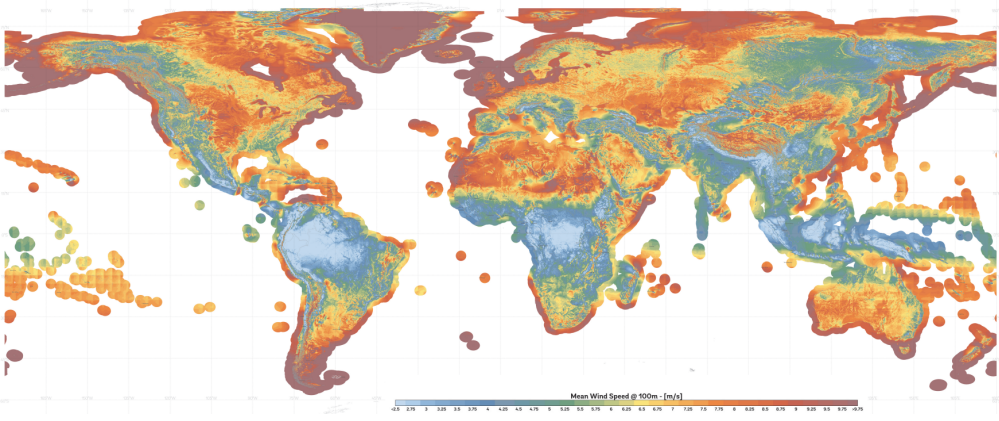


Figure: Mean wind speeds over the globe [3]

Geographical distribution

patterns in global wind atlas

Here are some pattern examples:

- Regions close to the Equator have low mean wind speeds.
- Regions close to the Poles have high mean wind speeds.
- In average, coastal regions have high mean wind speeds.
- In average, desert regions have high mean wind speeds.

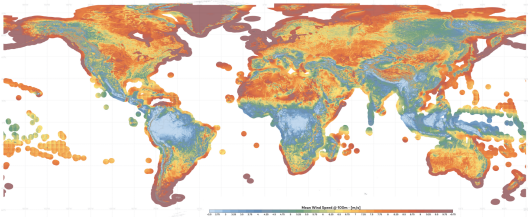


Figure: Mean wind speeds over the globe [3]

Terrain and height effects

boundary layer in fluid mechanics

The wind speeds (u) increase with height (z) relative to the surface and are reduced by the roughness of the terrain (z_0).

$$u(z) = u(z_1) \log\left(\frac{z - z_1}{z_0}\right) \qquad (1)$$

- $u(z)$: wind velocity at z height [m/s]
- $u(z_1)$: wind velocity at z_1 height [m/s]
- z_0: local roughness of surface [m]

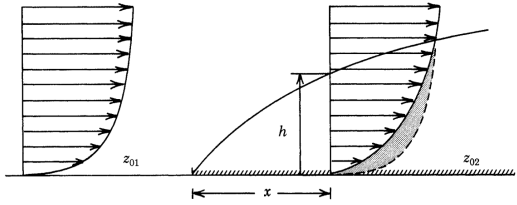


Figure: Wind velocity profile over two different terrain roughness ($z_{0,1} < z_{0,2}$) [4]

Terrain and height effects

boundary layer in fluid mechanics

Examples of roughness levels:

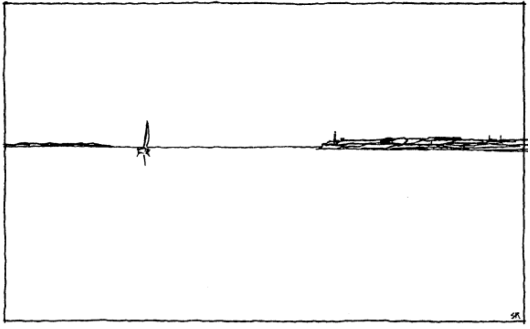


Figure: Sea, lakes and fjords: $z_0 = 0.0002$ m [4]

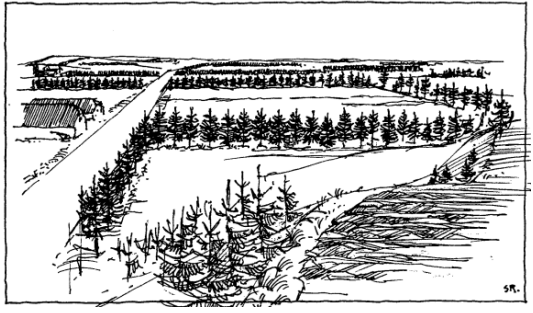


Figure: Urban, forest, and farm land with wind breaks: $z_0 = 0.4$ m [4]

Onshore vs. offshore

why moving offshore?

Offshore advantages

- Better wind quality: more energy from wind turbines.
- Less space restrictions: turbines can be larger.

Offshore disadvantages

- Expensive and complex: need offshore platforms and underwater cables.

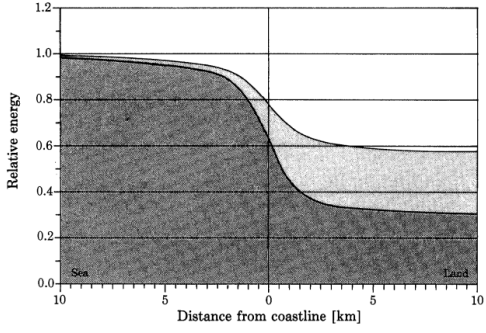


Figure: Mean energy density (50 m) for the Netherlands. Each line represents a roughness level. [4]

Activity

True or false

Consider the following statements:

1. The wind speeds for offshore regions are usually higher than for onshore regions for the same altitudes;
2. One of the main reasons wind speeds are higher for offshore conditions compared to onshore (at same heights) is the higher roughness of the sea;
3. The prevailing wind direction is southwest for whole northern hemisphere and northwest for the whole southern hemisphere;
4. The wind speed for lower altitudes is always higher than for higher altitudes because of air convection;
5. In general, regions close to the Equator have lower mean wind speed than other places in the globe.

It is possible to affirm:

a All statements are true

b Only 1 is true

c Only 4 is false

d 1 and 2 are true

e 2, 3 and 4 are false

Activity

True or false

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Variability

wind changes from time to time

A frequency plot represents the distribution of wind speeds.

Definition

Weibull function:

$$f(v) = \frac{k}{\lambda}\left(\frac{v}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{v}{\lambda}\right)^{k}\right) \quad (1)$$

- f(v): probability of v wind speed
- k: shape parameter
- λ: scale parameter

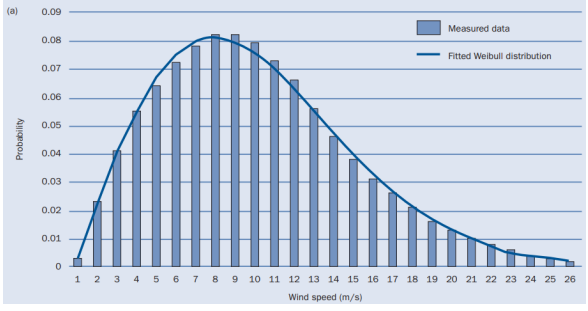


Figure: Example of wind speed distribution

Weibull parameters

Shape and scale

- The shape factor (k) relates to how much the distribution is concentrated in the average.
- The scale factor (λ) mainly relates to the average wind speed.

Example

Northern European climates often have shape factors k = 2 (i.e., Rayleigh distribution)

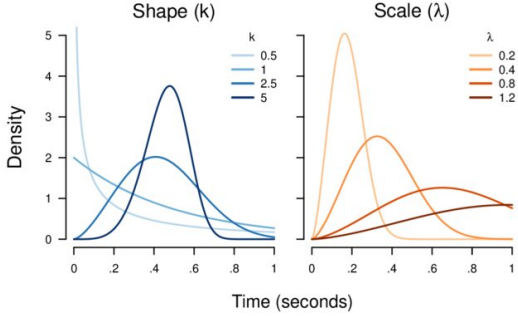


Figure: Effect of shape and scale parameters in Weibull distribution

Fluid flow and kinetic energy

calculating the total power in wind

$$P_{wind} = \frac{1}{2}\dot{m}v^2 \quad (2)$$

$$\dot{m} = \rho\dot{V} = \rho A v \quad (3)$$

- P_{wind}: Wind power [W]
- $\dot{m}$: Air mass flow [kg/s]
- v: wind velocity[a] [m/s]
- ρ: air density [kg/m^3]
- $\dot{V}$: volumetric flow rate [m^3/s]
- A: cross-section area [m^2]

[a] perpendicular to surface

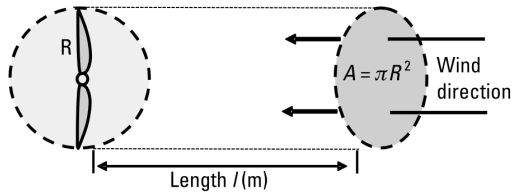


Figure: Kinetic energy of a fluid flow

Wind power density

how to measure the wind energy potential?

Definition

Wind power density (WPD):

$$\underbrace{WPD}_{P_w/A} = \frac{1}{2}\rho v^3 \tag{4}$$

Important

It is important to highlight that the cubic of the mean wind speed ($\bar{v}^3$) is different from the sum of the wind speed cubic ($\sum v_i^3$):

$$\bar{v}^3 \neq \frac{\sum_i^N v_i^3}{N}$$

Activity

Wind power density

Problem

In a hypothetical region, wind blows half of year with 10 m/s and the remaining time with 2 m/s. What is the yearly average wind power density of this region? Assume that air density is 1.2 kg/m^3

a. *4.8 W/m^2*
b. *129.6 W/m^2*
c. *259.2 W/m^2*
d. *302.4 W/m^2*
e. *604.8 W/m^2*

Activity

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Break

Check the time

Should we have a break (10 min.)?

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Betz limit

max conversion efficiency of wind power

Wind turbines can't convert all wind power into electricity because of mass and linear momentum conservation laws. The upper limit, called Betz limit, is 59.3 %.

$$P_t = C_p \overbrace{\frac{1}{2}\rho A v^3}^{P_{wind}} \quad (5)$$

$$C_p \leq \frac{16}{27} \quad (6)$$

- P_t: Turbine power [W]
- C_p: Power coefficient [-]

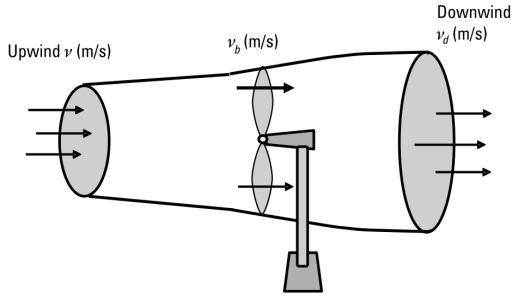


Figure: Simplified wind turbine balance [6]

Recap: rotation

angular velocity and torque

Angular velocity (ω) is a measure of how quickly an object rotates around an axis:

$$\omega = \frac{v}{R} \tag{7}$$

Torque is the product of the magnitude of the perpendicular component of the force (F) and the distance of the line of action of a force from the point around which it is being determined (R):

$$T = F \cdot R \tag{8}$$

Power generation (P) can be written as torque (T) times angular velocity (ω):

$$P = T \cdot \omega \tag{9}$$

Wake rotation

Turbines spin and angular momentum conserves

The Shimitz limit considers the conservation of angular momentum: the turbine spinning should create a spinning effect in the wind.

Definition

Tip speed ratio (λ_{TSP}):

$$\lambda_{TSP} = \frac{\omega R}{v} \qquad (10)$$

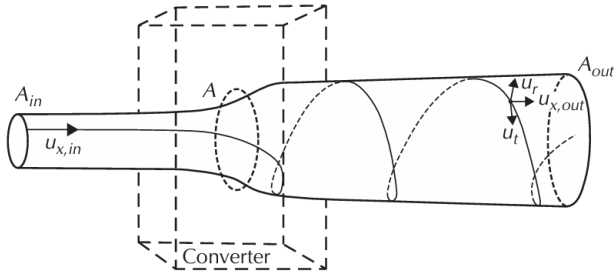


Figure: Schematic picture of a free-stream-flow converter or turbine.[8]

Wake rotation

Turbines spin and angular momentum conserves

- If λ_{TSP} is too low, air passes through the turbine without interacting with the blades.
- If λ_{TSP} is too high, the blade applies too much resistance to the wind flow.

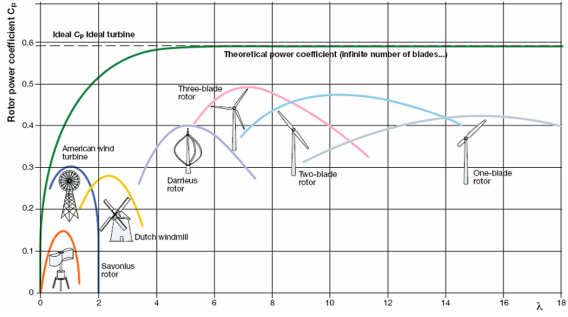


Figure: Power coefficient vs. tip speed ratio [9]

Activity

size matters

Problem

Two wind turbines operating at 8.5 m/s have the same efficiency in converting the total wind power. One turbine has 117 m of diameter and produces 1.814 MW. If the other turbine produces 2.472 MW, then what is its diameter?

a *125 m*

b *137 m*

c *145 m*

d *157 m*

e *165 m*

Activity

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a *125 m*

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Power curve

Turbine power vs. wind speed

Definition

- Rated power (P_{rated}): The design power generation.
- Cut-in speed (v_{in}): Min. wind speed to generate power.
- Shutdown / cut-out speed (v_{out}): Max. wind speed to generate power.
- Rated speed (v_{rated}): Min. wind speed that gives the design power generation.

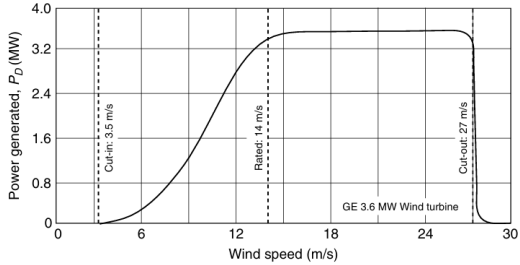


Figure: Wind turbine power-speed curve [7]

Activity

Energy losses

Problem

What percentage of wind power is not converted into electricity at rated speed? The wind turbine has a 104 m diameter and $\rho_{air} = 1.2\ [kg/m^3]$.

a *13%*

b *26%*

c *74%*

d *87%*

e *94%*

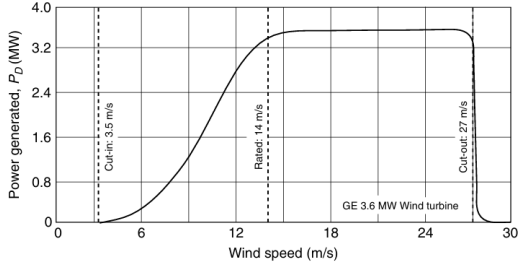


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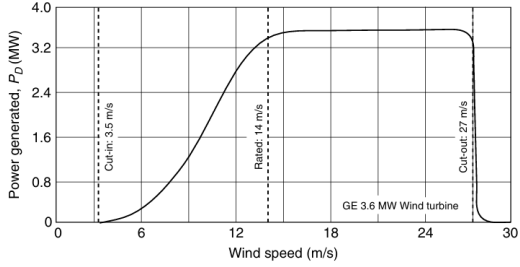


Figure: Wind turbine power-speed curve [7]

Annual energy production (AEP)

How much energy a wind turbine produce in a year?

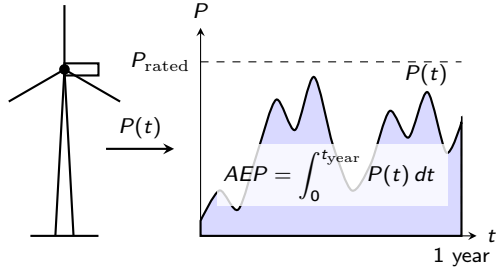
AEP is the total energy produced in a year:

$$AEP = \sum_{t} P_t \Delta t \qquad (10)$$

It is often given indirectly from the capacity factor:

$$CF = \frac{AEP}{P_{rated} \times t_{year}} \qquad (11)$$

In which, t_{year} is usually assumed as 8760 hours.



Activity

Capacity factor and annual energy production

Problem (2023 exam)

A wind turbine with rated power of $P = 15$ MW and rotor diameter of $D = 236$ m is installed in an offshore wind farm, where the capacity factor (CF) is expected to be CF $= 0.55$. What is the expected Annual Energy Production (AEP) of the turbine?

a *20-50 GWh/y*

b *50-80 GWh/y*

c *80-110 GWh/y*

d *110-140 GWh/y*

e *140-170 GWh/y*

Activity

Capacity factor and annual energy production

Problem (2023 exam)

A wind turbine with rated power of $P = 15$ MW and rotor diameter of $D = 236$ m is installed in an offshore wind farm, where the capacity factor (CF) is expected to be CF $= 0.55$. What is the expected Annual Energy Production (AEP) of the turbine?

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d *110-140 GWh/y*

e *140-170 GWh/y*

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Aerodynamics

friction and pressure differences

Fluid flow around an object can exert a force that is commonly decomposed into two components:

Definition

- Lift is the force component perpendicular to the flow
- Drag is the force component parallel to the flow

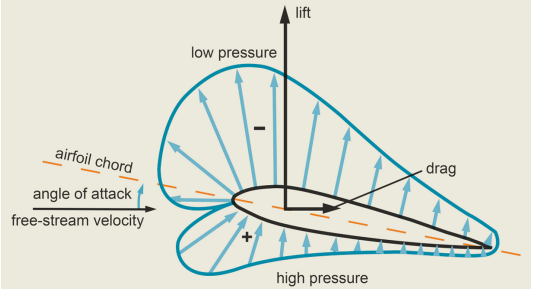


Figure: Pressure in an airfoil [10]

Typical wind turbine

Three blades, horizontal axis, propeller type

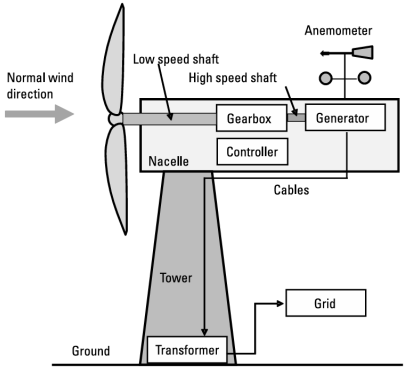


Figure: Wind turbine parts [6]

The pitch system controls the angle of the blades, and the yaw system controls the orientation.

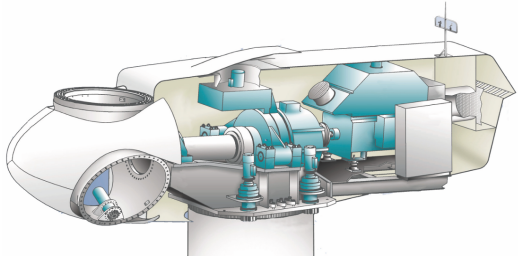


Figure: Details of the General Electric GE 1.5 ls [10]

Offshore foundations

fixed and floating options

Fixed foundations are considered economically viable for waters up to 60 m deep. 80% of the world's offshore wind potential is above that.

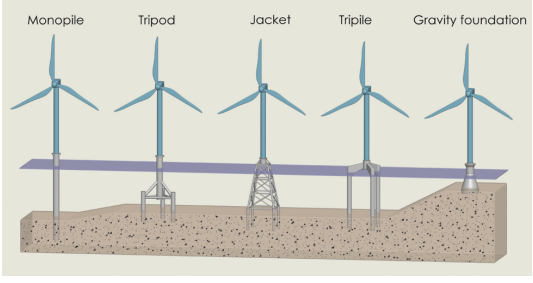


Figure: Fixed foundations for offshore wind turbines [10]

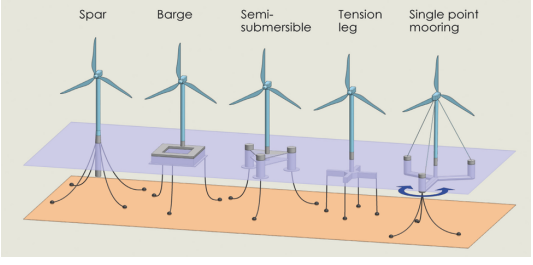


Figure: Floating foundations for offshore wind turbines [10]

Technology trends

going big and placement challenges

Going big gets better wind and reduces specific costs, but has structural challenges.

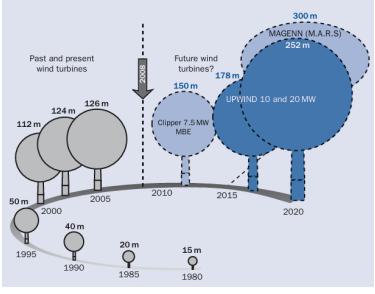


Figure: Size trend for wind turbines [11]

Concerns about interference in radar systems limit the available locations for offshore wind.

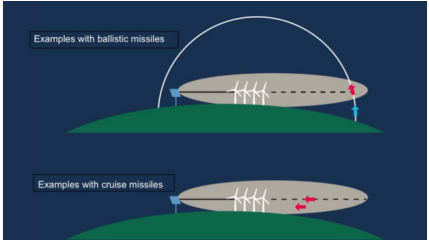


Figure: Example of radar interference of offshore wind [12]

Activity

Let's vote!

Problem

Are environmental and national security concerns sufficient reasons to stop an otherwise viable offshore wind project?

a *Yes*

b *No*

Note

You will have to defend your position afterwards.

Muddy points and key insights

Feedback for your learning

Here is the chance of you get feedback on your learning.

The quiz will be open only TODAY.

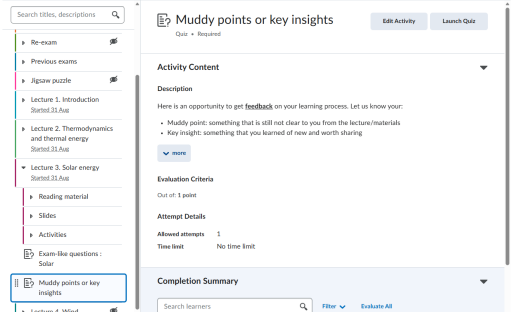


Figure: A screenshot capture of your DTU Learn

- U.S. Global Change Research Program: Climate Science Special Report, Fourth National Climate Assessment (NCA4), Volume I, chapter 5.1. https://en.wikipedia.org/wiki/Hadley_cell
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